

An Investigation of a Notion of „Variable Forgetting“ that Minimizes Truth Values

Bachelor's Thesis

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Abstract

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1 Motivation

Why ought an agent forget acquired knowledge at all? Wouldn't that be fatal? Should we just forget that the Earth is a sphere? Is the result a step forward or backward? We presuppose that Truth or what is socially valid as Truth changes in the context of time and history. Practically, knowledge or facts are indeed compelling but not dogmatic and can be falsified by the progress of scientific practice. Knowledge changes, can be replaced by other knowledge, or become irrelevant. Whether it's about knowledge representation, reasoning, belief change, or multi-agent systems, in Artificial Intelligence, there are many examples where knowledge modeling is a crucial component. Thus, it's clear that it's advantageous not only to acquire knowledge but also to be able to eliminate it. We don't just want to model strict knowledge but also views, values or opinions. As is well known, these change in the context of time and the social circumstance in which they are carried. Certain views become irrelevant and simply forgotten. Furthermore, there are purely material difficulties, as computing complex queries to a knowledge base or drawing conclusions by artificial agents requires high computational power. Filtering out irrelevant knowledge also means reducing complexity, from which many areas of AI would technically benefit. Another perspective is that of focus. Knowledge changes depending on what is considered. If certain connections are not seen, the whole shape changes. What is deemed relevant also determines knowledge. Here we have a rather indirect or presupposed perspective for our topic. What is irrelevant is consequently also forgotten. To this end there are a few approaches to forgetting in logic yet it is not a well researched topic. One can come up with many ways of what forgetting even means and how it should work. We will concentrate on the variable forgetting in propositional logic. In the laws of thought [Boo21] George Boole denotes the so called "elimination of the middle terms" which can be seen as an early version of variable forget in the field. The Forget operation that we want to build upon, goes back to the paper Forget It! [LR94] by Lin and Reiter in 1994 which takes the approach to make the variable irrelevant. Their notion is defined by having two interpretations that totally agree about anything except the truthvalues of variable to forget, hence it is irrelevant. We will define a variation of this operation, which we call the SkipForget operation and that we want to investigate in this thesis.

1.1 Preliminaries

Let L be a propositional language. We denote set of propositional variables by V . Sentences or Formulas in L over V inductively defined over the common logical connectives ($\neg, \wedge, \vee, \rightarrow, \leftrightarrow$) as well as the logical constants *true* and *false*. For every propositional variable $a \in V$, a is sentence. If F is a sentence, then so is $\neg F$. If F and G are sentences, then so are $(F \wedge G)$, $(F \vee G)$, $(F \rightarrow G)$, and $(F \leftrightarrow G)$. For instance, $F = (\neg(a \wedge b) \vee (c \rightarrow d))$ is a sentence. As stated in [LR94] or [LLM03], we define variable forgetting inductively as follows:

Definition 1 (Skeptical Variable Forgetting) Let F be a formula, p an atomic variable and S as Set of Variables. The skeptical forgetting of p in F $SkepForget(F, p)$, respectively, the skeptical forgetting of S in F $SkepForget(F, S)$, is given inductively by:

- $SkepForget(F, \emptyset) = F$
- $SkepForget(F, \{p\}) = F[p/\top] \wedge F[p/\perp]$
- $SkepForget(F, p) = SkepForget(F, \{p\})$
- $SkepForget(F, S \cup \{p\}) = SkepForget(SkepForget(F, \{p\}), S)$

For some formulas F , this operation of variable elimination can lead to a notable reductive or drastic results. We start by considering an example where *Forget* behaves quite appropriate. Let a sentence F_1 be $a \wedge b$ and we choose to forget about b :

$$\begin{aligned} Forget(F_1, b) &= F_1[b/\top] \vee F_1[b/\perp] \\ &\equiv ((a \wedge \top) \vee (a \wedge \perp)) \\ &\equiv (a \vee \perp) \\ &\equiv a \end{aligned}$$

We can see, as shown above, that $Forget(F_1, b)$ is equivalent to a which seems to be the intuitive result when thinking of forgetting about b in F_1 . Next, we consider forgetting of b in the formula $F_2 = a \vee b$

$$\begin{aligned} Forget(F_2, b) &= F_2[b/\top] \vee F_2[b/\perp] \\ &\equiv ((a \vee \top) \vee (a \vee \perp)) \\ &\equiv (\top \vee a) \\ &\equiv \top \end{aligned}$$

We observe that $Forget(F_2, b)$ is *tautological*, which can be seen as a rather drastic result when we choose to forget b only. A variation of a forget operation could be defined as follows:

Definition 2 (Skeptical Variable Forgetting) Let F be a formula, p an atomic variable and S as Set of Variables. The skeptical forgetting of p in F $SkepForget(F, p)$, respectively, the skeptical forgetting of S in F $SkepForget(F, S)$, is given inductively by:

- $SkepForget(F, \emptyset) = F$
- $SkepForget(F, \{p\}) = F[p/\top] \wedge F[p/\perp]$
- $SkepForget(F, p) = SkepForget(F, \{p\})$
- $SkepForget(F, S \cup \{p\}) = SkepForget(SkepForget(F, \{p\}), S)$

The idea is to use conjunction instead of disjunction. The consequences for our previous examples are the following:

$$\begin{aligned}
SkepForget(F_1, b) &= F_1[b/\top] \vee F_1[b/\perp] \\
&\equiv ((a \wedge \top) \wedge (a \wedge \perp)) \\
&\equiv (a \wedge \perp) \\
&\equiv \perp
\end{aligned}$$

$$\begin{aligned}
SkepForget(F_2, b) &= F_2[b/\top] \vee F_2[b/\perp] \\
&\equiv ((a \vee \top) \wedge (a \vee \perp)) \\
&\equiv (\top \wedge a) \\
&\equiv a
\end{aligned}$$

We observe that our alternative operation, at least in these examples, doesn't necessarily entail less drastic results. We obtain a contradiction in the first case and in the second case, our equivalent to a . It suggests a duality between the two operations.

2 SyntacticAnalysis

To explore the properties of our *SkepForget* operation, we need to identify the space or dimensions in which the function operates. While introductory examples provided in the previous section shed light on the primary issue, it remains crucial to investigate the behavior of *SkepForget* with more complex sentences. Do the outcomes predominantly correlate with the top layer of connectives in the syntax tree? What influence do deeper nested expressions exert on the final result?

To draw valuable conclusions it is necessary to select examples that can proof generality while being specific enough to not lose any valuable properties and characteristics. How to identify such examples? What criteria are essential for their selection?

From the syntactical perspective we have sketched several dimensions for comparison: We need to examine the behavior of the two forget operations in relation — are they dual in nature or not? We need to consider complexity of sentences such as negation and multiple binary connectives with deeper nesting. Comparing the forgetting of variables at different depths, for instance, for a sentence such as (a or b or c) we have three options.

Lets start our investigation with the following sentences:

$$\begin{array}{llll}
F_{\alpha 1} &\equiv a & Forget(F_{\alpha 1}, a) &\equiv \top & SkepForget(F_{\alpha 1}, a) &\equiv \perp \\
F_{\alpha 2} &\equiv \neg a & Forget(F_{\alpha 2}, a) &\equiv \top & SkepForget(F_{\alpha 2}, a) &\equiv \perp
\end{array}$$

In $F_{\alpha 1}$ and $F_{\alpha 2}$ we forget the whole sentence itself where $F_{\alpha 2}$ is negation of $F_{\alpha 1}$. In both cases the result is

top for *Forget* and

bot for *SkepForget* as if it would characterize the function fundamentally. Lets see what happens with more complex sentences.

$$\begin{array}{lll}
F_{\beta 1} \equiv (a \wedge b) & Forget(F_{\beta 1}, a) \equiv b & SkepForget(F_{\beta 1}, a) \equiv \perp \\
F_{\beta 2} \equiv \neg(a \wedge b) & Forget(F_{\beta 2}, a) \equiv \top & SkepForget(F_{\beta 2}, a) \equiv \neg b \\
F_{\beta 3} \equiv (a \vee b) & Forget(F_{\beta 3}, a) \equiv \top & SkepForget(F_{\beta 3}, a) \equiv b \\
F_{\beta 4} \equiv \neg(a \vee b) & Forget(F_{\beta 4}, a) \equiv \neg b & SkepForget(F_{\beta 4}, a) \equiv \perp
\end{array}$$

Here we can see that $Forget(F_{\beta 1}, a) \equiv \neg(SkepForget(F_{\beta 2}, a))$

With $F_{\beta 1} \equiv \neg F_{\beta 2}$ we get $Forget(F_{\beta 1}, a) \equiv \neg SkepForget(\neg F_{\beta 1}, a)$ $\neg F_{\beta 1} \equiv F_{\beta 2}$ and $Forget(\neg F_{\beta 2}, a) \equiv \neg SkepForget(F_{\beta 2}, a)$

More generally we can establish a relation that: $Forget(F, a) \equiv \neg SkepForget(\neg F, a)$ and $Forget(\neg F, a) \equiv \neg SkepForget(F, a)$

Lets focus more on the logical connectives:

$$\begin{array}{lll}
F_{\gamma 1} \equiv (a \wedge b) & Forget(F_{\gamma 1}, a) \equiv b & SkepForget(F_{\gamma 1}, a) \equiv \perp \\
F_{\gamma 2} \equiv (a \vee b) & Forget(F_{\gamma 2}, a) \equiv \top & SkepForget(F_{\gamma 2}, a) \equiv b
\end{array}$$

In $F_{\beta 1}$ and $F_{\beta 2}$ we forget only one variable of a binary connective. Since logical connectives are commutative, forgetting the other variable would not give us new information. We observe a dualistic behavior. Our forget operations deliver either truth values or the remaining variable b of the original sentence. A necessary condition seems to be the connective itself. We can observe that Forget with $\wedge$ gives us $\perp$ where in combination with $\vee$ we receive b . For $SkepForget$ its the other way around, also the truth value is $\top$ which is the opposite Bottom line, with the 4 results, we loose exactly the variable, which we expect to loose. And in the other two cases we loose more information that we expect Lets see if this holds true if we look are more complex/longer sentences:

$$\begin{array}{lll}
F_{\delta 1} \equiv (a \wedge (c \wedge d)) & Forget(F_{\delta 1}, a) \equiv (c \wedge d) & SkepForget(F_{\delta 1}, a) \equiv \perp \\
F_{\delta 2} \equiv (a \wedge (c \vee d)) & Forget(F_{\delta 2}, a) \equiv (c \vee d) & SkepForget(F_{\delta 2}, a) \equiv \perp \\
F_{\delta 3} \equiv (a \vee (c \wedge d)) & Forget(F_{\delta 3}, a) \equiv \top & SkepForget(F_{\delta 3}, a) \equiv (c \wedge d) \\
F_{\delta 4} \equiv (a \vee (c \vee d)) & Forget(F_{\delta 4}, a) \equiv \top & SkepForget(F_{\delta 4}, a) \equiv (c \vee d)
\end{array}$$

If we add more complexity via one additional connective. It seems that, due to the inductive or recursive structure of propositional logic we still get the same dualistic result as in the previous example. As in the previous example, in half of the cases we loose exactly what we expect, and in the other half we loose more information. Just to be on the safe side, lets see if it still holds when we add one more variable

$F_{\theta 1} \equiv ((a \wedge b) \wedge (c \wedge d))$	$Forget(F_{\theta 1}, a) \equiv (b \wedge (c \wedge d))$	$SkepForget(F_{\theta 1}, a) \equiv \perp$
$F_{\theta 2} \equiv ((a \wedge b) \wedge (c \vee d))$	$Forget(F_{\theta 2}, a) \equiv (b \wedge (c \vee d))$	$SkepForget(F_{\theta 2}, a) \equiv \perp$
$F_{\theta 3} \equiv ((a \wedge b) \vee (c \wedge d))$	$Forget(F_{\theta 3}, a) \equiv (b \vee (c \wedge d))$	$SkepForget(F_{\theta 3}, a) \equiv (c \wedge d)$
$F_{\theta 4} \equiv ((a \wedge b) \vee (c \vee d))$	$Forget(F_{\theta 4}, a) \equiv (b \vee (c \vee d))$	$SkepForget(F_{\theta 4}, a) \equiv (c \vee d)$
$F_{\theta 5} \equiv ((a \vee b) \wedge (c \wedge d))$	$Forget(F_{\theta 5}, a) \equiv (c \wedge d)$	$SkepForget(F_{\theta 5}, a) \equiv (b \wedge (c \wedge d))$
$F_{\theta 6} \equiv ((a \vee b) \wedge (c \vee d))$	$Forget(F_{\theta 6}, a) \equiv (c \vee d)$	$SkepForget(F_{\theta 6}, a) \equiv (b \wedge (c \vee d))$
$F_{\theta 7} \equiv ((a \vee b) \vee (c \wedge d))$	$Forget(F_{\theta 7}, a) \equiv \top$	$SkepForget(F_{\theta 7}, a) \equiv (b \vee (c \wedge d))$
$F_{\theta 8} \equiv ((a \vee b) \vee (c \vee d))$	$Forget(F_{\theta 8}, a) \equiv \top$	$SkepForget(F_{\theta 8}, a) \equiv (b \vee (c \vee d))$

Surprisingly we dont get exactly the same behaviour as before. The dualistic over-all picture remains. But for $F_{\theta 5}$ and $F_{\theta 6}$ on the Forget side and $F_{\theta 3}$ and $F_{\theta 4}$ on the SkepForget side the result is structurally different from what we expected compared to the previous example. Because, as in the previous example, in half of the cases we loose exactly what we want to loose but in the other half it is more differentiated. We loose some but we dont loos as much as in the previous. But we made a little mistake here: Imagine a from the previous example is now replaced with $(a \circ b)$, we now forget only a and not $(a \circ b)$ wich we should in order to make the same move as before but we can see that within the half of the cases in wich we loose information again we can divide this in half and observe that we only lost half of that in drastic way, meaning reuslt is only truth values As we can only forget atomic variables, we cant make this comparison as we would need to forget $(a \circ b)$. We now would need to look at forgetting different variables in same sentences: note: also a is now deeper nested vs before it was top layer

For a sentence $a \wedge (b \vee c)$ we have 3 options, but only 2 layers in depth, considering commutativity of logical connectives we can reduce this to 2 options.

$F_{\phi 1} \equiv (a \wedge (b \wedge c))$	$Forget(F_{\phi 1}, a) \equiv (b \wedge c)$	$SkepForget(F_{\phi 1}, a) \equiv \perp$
$F_{\phi 2} \equiv (a \wedge (b \vee c))$	$Forget(F_{\phi 2}, a) \equiv (b \vee c)$	$SkepForget(F_{\phi 2}, a) \equiv \perp$
$F_{\phi 3} \equiv (a \vee (b \wedge c))$	$Forget(F_{\phi 3}, a) \equiv \top$	$SkepForget(F_{\phi 3}, a) \equiv (b \wedge c)$
$F_{\phi 4} \equiv (a \vee (b \vee c))$	$Forget(F_{\phi 4}, a) \equiv \top$	$SkepForget(F_{\phi 4}, a) \equiv (b \vee c)$
$F_{\phi 1} \equiv (a \wedge (b \wedge c))$	$Forget(F_{\phi 1}, c) \equiv (a \wedge b)$	$SkepForget(F_{\phi 1}, c) \equiv \perp$
$F_{\phi 2} \equiv (a \wedge (b \vee c))$	$Forget(F_{\phi 2}, c) \equiv a$	$SkepForget(F_{\phi 2}, c) \equiv (a \wedge b)$
$F_{\phi 3} \equiv (a \vee (b \wedge c))$	$Forget(F_{\phi 3}, c) \equiv (a \vee b)$	$SkepForget(F_{\phi 3}, c) \equiv a$
$F_{\phi 4} \equiv (a \vee (b \vee c))$	$Forget(F_{\phi 4}, c) \equiv \top$	$SkepForget(F_{\phi 4}, c) \equiv (a \vee b)$

taking on the perspective from the variable to forget we can see that half of the cases we loose only what we expect.

a is in the top layer, here will get themost drastic results where the deeper nested variable c is were we get the less drastic results, we loose less information

It seems that we loose information.

2.1 Behavior of Truth Values in Abstract Syntax Trees

The behavior of truth values presents notably insights when considering their interactions with logical connectives. We know that $\top \wedge \perp \equiv \perp$ and $\top \vee \perp \equiv \top$.

To investigate the behaviour abstract syntax trees are a great way to visualize sentences. To get a good overview we will try any permutation of connectives over a certain variable amount n which is $2^{(n-1)}$ possibilities.

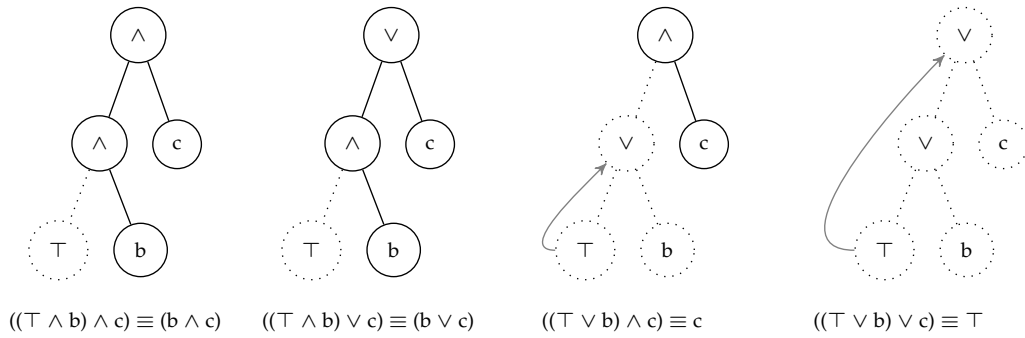


Figure 1: $\top$ propagation

In figure 3 we can observe the propagation of truth values, in this case $\top$. Starting from the leafes @ $\top$ effectively "absorbs" all siblings in conjunction with $\vee$ and stops propagating at $\wedge$.

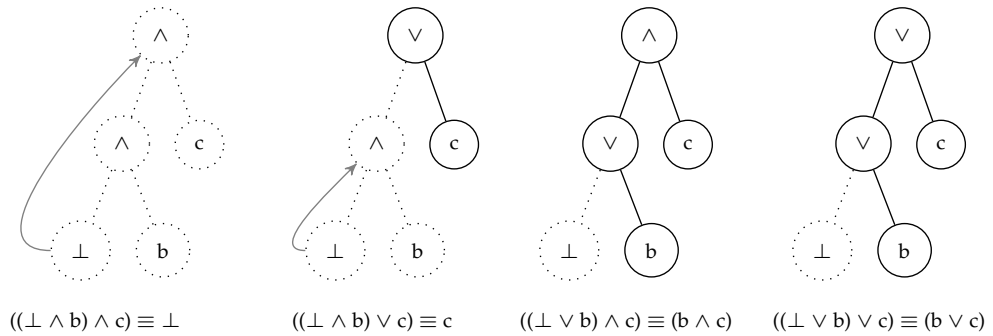


Figure 2: $\perp$ propagation

Figure 2 shows, dual to 3, how $\perp$ "annihilates" all siblings when paired with a logical $\wedge$ and halts at $\vee$. This dynamic interaction implies that truth values possess the capability to selectively "eliminate" variables and entire propositions.

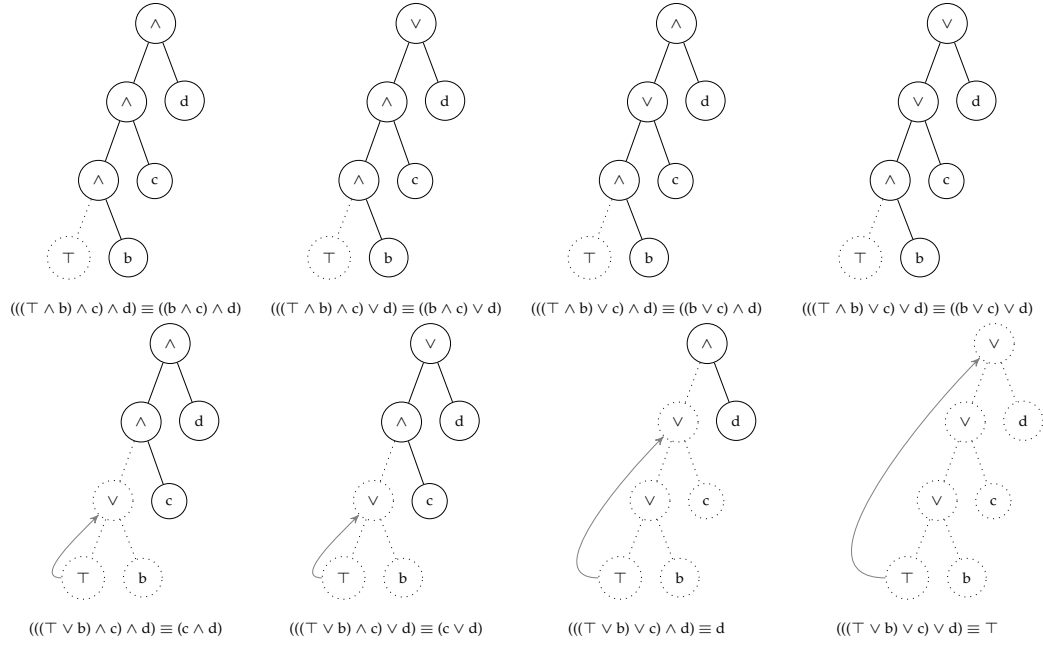


Figure 3: T propagation

Lets say V is the Set of variables in a Sentence, and $L \in \mathbb{N} \geq 0$ is the layer in the syntax tree. We can denote the possible amount of truth values for any sentence over all possible connectives as $TVA = 2^{|V|-1}/2^L$, the partially lost information as $PL = (2^{|V|-1}/2) - TVA$ and the fully kept information as $K = (2^{|V|-1}/2)$

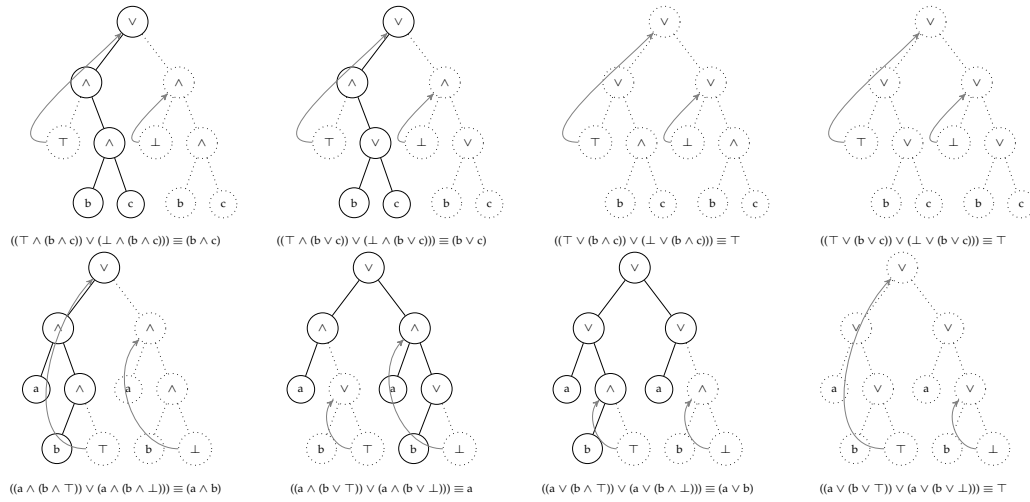


Figure 4: ddddd

3 Semantic Investigation

In [LLM03], the authors denote two semantic functions, Force and Switch, as follows: Given in interpretation w and a literal l , we let $\text{Force}(w, l)$ denote the interpretation that gives the same truth value as w for all variables except the variable of l , and such that $\text{Force}(w, l) \models l$. Meaning $\text{Force}(w, l)$ is the interpretation that satisfying l that is closest to w , If $w \models l$ then $\text{Force}(w, l) = w$.

$\text{Switch}(w, l)$ denotes the interpretation that maintains the same truth value as w for all variables except x , just like Force, but assigns to x the opposite truth value given by w .

3.1 Properties of Forget

In [LR94] the authors state the following properties of Forget:

- $T \models \text{forget}(T, p)$
- For any sentence ϕ , $T \models \phi \downarrow p \text{ iff } \text{forget}(T; p) \models \phi \downarrow p$ (works for FOL, not pl ???)

[EKI19]

- $M_1 \sim_p M_2$, M_1 and M_2 agree on all variables except possibly p

In [LLM03] the authors state the following properties of Forget:

- Proposition 14: The set of models of $\text{ForgetLit}(\Sigma, \{l\})$ can be expressed as:
Corollary 5: $\text{Mod}(\text{ForgetLit}(\Sigma, \{l\})) = \text{Mod}(\Sigma) \cup \{\text{Force}(w, -l) \mid w \models \Sigma\} = \{w \mid \text{Force}(w, l) \models \Sigma\}$
- $\text{Mod}(\text{Forget}(F, x)) = \text{Mod}(F) \cup \{\text{Switch}(w, x) \mid w \models F\}$

$$\text{Int}_{\Sigma_2}(\text{Forget}(F, a)) = \text{Mod}_{\Sigma_1}(a)$$

$$\forall w : w \models a, w(\text{Forget}(F, a)) = w(F)$$

$$\forall w : w \not\models a, w(\text{Forget}(F, a)) = \text{Force}(w, A)(F)$$

3.2 Properties of SkepForget

Proposition: $Mod(SkepForget(F, x)) = Mod(F) \cap \{Switch(w, x) | w \models F\}$

Let F be a propositional sentence, over a signature $\Sigma = \{a, b\}$, a propositional variable $x \in \Sigma$ and w be an interpretation.

Using structural induction, we can prove that the proposition holds. We consider the following cases:

1. **Base Case:** F is a propositional atomic variable.

$$Mod_{\Sigma_1}(a) \equiv \begin{array}{|c|c|} \hline a & a \\ \hline 1 & 1 \\ \hline \end{array}$$

$$Switch(w, a) | w \models a \equiv \begin{array}{|c|c|} \hline a & a \\ \hline 0 & 0 \\ \hline \end{array}$$

$$Mod_{\Sigma_1}(SkepForget(a)) \equiv \emptyset$$

$$Mod(F)_{\Sigma_1} \cap \{Switch(w, a) | w \models a\} \equiv \emptyset$$

with $\Sigma_1\{a\}$

2. **Negation Case:**

$$Mod_{\Sigma_1}(\neg a) \equiv \begin{array}{|c|c|} \hline a & \neg a \\ \hline 0 & 1 \\ \hline \end{array}$$

$$Switch(w, a) | w \models \neg a \equiv \begin{array}{|c|c|} \hline a & \neg a \\ \hline 1 & 0 \\ \hline \end{array}$$

$$Mod_{\Sigma_1}(SkepForget(\neg a)) \equiv \emptyset$$

$$Mod(F)_{\Sigma_1} \cap \{Switch(w, a) | w \models \neg a\} \equiv \emptyset$$

with $\Sigma_1\{a\}$

3. **$\vee$ Case:**

$$\begin{array}{lcl}
Mod_{\Sigma_2}((a \vee b)) & \equiv & \begin{array}{|c|c|c|} \hline a & b & (a \vee b) \\ \hline 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ a & b & (a \vee b) \\ \hline \end{array} \\
Switch(w, a)|w \models (a \vee b) & \equiv & \begin{array}{|c|c|c|} \hline 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \\ a & b & ((\top \vee b) \wedge (\perp \vee b)) \\ \hline \end{array} \\
Mod_{\Sigma_2}(SkepForget((a \vee b))) & \equiv & \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ 0 & 1 & 1 \\ a & b & (a \vee b) \\ \hline \end{array} \\
Mod(F)_{\Sigma_2} \cap \{Switch(w, a)|w \models (a \vee b)\} & \equiv & \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ 0 & 1 & 1 \\ \hline \end{array} \\
\text{with } \Sigma_2\{a, b\}
\end{array}$$

4. $\wedge$ **Case:**

$$\begin{array}{lcl}
Mod_{\Sigma_2}((a \wedge b)) & \equiv & \begin{array}{|c|c|c|} \hline a & b & (a \wedge b) \\ \hline 1 & 1 & 1 \\ \hline \end{array} \\
Switch(w, a)|w \models (a \wedge b) & \equiv & \begin{array}{|c|c|c|} \hline a & b & (a \wedge b) \\ \hline 0 & 1 & 0 \\ \hline \end{array} \\
Mod_{\Sigma_2}(SkepForget((a \wedge b))) & \equiv & \emptyset \\
Mod(F)_{\Sigma_2} \cap \{Switch(w, a)|w \models (a \wedge b)\} & \equiv & \emptyset
\end{array}$$

with $\Sigma_2\{a, b\}$

a	b	$\neg a$	$((\top \wedge b) \wedge (\perp \wedge b))$	$((\top \vee b) \wedge (\perp \vee b))$	$(\neg \top \wedge \neg \perp)$	$(a \wedge b)$	$(a \vee b)$
1	1	0	0	1	0	1	1
1	0	0	0	0	0	0	1
0	1	1	0	1	0	0	1
0	0	1	0	0	0	0	0

reference

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